



THE PLUMAGE AND COLOURATION OF AN ENANTIORNITHINE BIRD FROM THE EARLY CRETACEOUS OF CHINA

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Abstract: Brilliant colour displays and diverse feather morphologies that are often sexual ornaments are common throughout much of extant Avialae. Here we describe a new basal enantiornithine bird specimen recovered from the Early Cretaceous Jiufotang Formation of Liaoning Province in northeastern China. We present new information on the plumage of Bohaiornithidae as well as the first detailed colour reconstruction of an enantiornithine bird. The new specimen retains subadult skeletal characteristics, including periosteal pitting of the long bone epiphyses and unfused elements, while also preserving plumage evidence consistent with sexual maturity at the time of death. Exceptionally-preserved feathers cover the body, including elongate crown feathers,

body contour feathers, asymmetrically-veined wing primaries, an alula and two elongate rachis-dominated rectrices that may have been sexual ornaments. The crown, neck, and body contour feathers retain elongate melanosome morphologies associated with weakly iridescent colouration in extant feathers. We provide additional evidence of preserved melanin using Raman spectroscopy; a rapid, non-destructive chemical technique. The new specimen provides data on skeletal ontogeny in the Bohaiornithidae as well as evidence for intraspecific communication functions of plumage.

Key words: Enantiornithes, bird, sexual selection, melanosome, feather colour, Bohaiornithidae.

THE exceptionally-preserved dinosaurs from the Early Cretaceous Jehol Lagerstätte of northeastern China have greatly contributed to our understanding of avian evolution and ecology. The Jehol biota preserves a highly diverse assemblage of avian and non-avian dinosaurs, in many cases including the integument, allowing for inferences concerning avian flight (Zhou 2004; Norell & Xu 2005; Clarke & Middleton 2008; Zhou *et al.* 2008; Heers & Dial 2012; Xu *et al.* 2014) and colouration (Li *et al.* 2010, 2012; Huang *et al.* 2016). Here, we describe a new bohaiornithid bird specimen from the Early Cretaceous Jiufotang Formation with remarkably preserved feathers and present, to our knowledge, the first reconstruction of the colour of an enantiornithine bird. Bohaiornithidae (including *Bohaiornis*, *Longsunguis*, *Parabohaiornis*, *Shenqiornis*, *Sulcavis* and *Zhouornis*) is the most speciose recognized clade of enantiornithine birds. This derived group, of which *Shenqiornis* is the oldest, spanned across at least five million years (120–125 Ma; Wang *et al.* 2014a). While potential diets and dominant habitats have been inferred for some bohaiornithids, little is known

about their ecology (Li *et al.* 2014a; Wang *et al.* 2014a). Feather colouration is relevant to numerous aspects of extant avian ecology, including sexual selection, camouflage and other inter and intraspecific communications (McGraw 2006). Poor preservation of the plumage in previously described Bohaiornithidae means that nothing is known of their plumage colour. CUGB P1202 is a young individual that retains the best preservation of plumage of any bohaiornithid, including preserved melanosomes with chemically preserved melanin, exceptionally detailed primary remiges, an alula, elongate crown feathers, contour feathers of the nape and body, numerous short rectrices and paired elongate, rachis-dominated, rectrices.

MATERIAL AND METHOD

Specimen and sampling

Specimen CUGB P1202 (CUGB, China University of Geosciences, Beijing) consists of a part and counterpart.

It was collected from the Early Cretaceous Jiufotang Formation, near Lamadong Village, Jianchang County, Liaoning Province, China. Bone and feather morphology was measured from high-resolution digital images using ImageJ (<http://rsbweb.nih.gov/ij/>). Using previously described techniques (Li *et al.* 2010), we removed samples measuring approximately 1 mm in length from four different exceptionally-preserved feathers of the CUGB P1202 primary slab (Peteya *et al.* 2016, fig. S1) for scanning electron microscopy (SEM) and Raman spectroscopy. We removed one sample for SEM and a separate, adjacent sample for Raman spectroscopy, from feathers of the crown, the nape, the right wing next to the ulna, and contour feathers next to the distal right tibia and proximal tarsometatarsus. As a control, we also sampled the grey-coloured matrix in which the fossil is preserved, white-coloured matrix above the fossil, and white-coloured matrix between the bones and the preserved feathers. Sampling locations within the white matrix include near the cranial and caudal cervical vertebrae, below the right dentary and above the left humeral head on the counter slab (Peteya *et al.* 2016, fig. S1). As in the case of the fossil samples, we took separate samples from each targeted area of the matrix for Raman spectroscopy and SEM. X-ray computed tomography images of the specimen were acquired using a high-resolution X-ray CT scanner (Nikon XT H 320 LC) at The China University of Geosciences. We generated 80 slices each with an image resolution of 2000×2000 pixels along the coronal axis of the primary slab using a voltage of 180 kV, a current of 118 μ A, no filter, and a 0.075211 mm voxel size.

Scanning electron microscopy

Before imaging, we placed the fossil and matrix samples on carbon tape and sputter-coated them for three minutes in gold–palladium using a Polaron E5000 Sputter-Coater (Quorum Technologies). We used a JSM-7401 Field Emission Scanning Electron Microscope (JEOL Solutions for Innovation) to image the samples at 7 kV and a 7 mm working distance.

Based on chemical and morphological similarities to melanosomes in extant bird feathers (see below), elongate three-dimensional granules and impressions preserved in the feathers of CUGB P1202 are interpreted as melanosomes. We measured the size and shape of melanosomes of both preservational modes and compared these parameters to melanosome morphology in extant organisms using previously described techniques (Clarke *et al.* 2010; Li *et al.* 2010, 2012, 2014b). Lengths (long axis) and widths (short axis) of complete three-dimensionally preserved melanosomes and melanosome impressions were

measured from SEM images using ImageJ. These values were then used to calculate the aspect (length:width) ratio of each measured melanosome.

We used quadratic canonical discriminant analysis (CDA) in JMP Pro 11 (SAS Institute Inc.) to predict the colour of CUGB P1202. This standard method has previously been used to predict the colour of extant avian feathers with 82% accuracy (Li *et al.* 2012). The size (length and width) and shape (aspect ratio) of melanosomes preserved in CUGB P1202 were compared to those from extant bird feathers, which can be separated into five categories based on colour and morphology: black, brown, grey, iridescent and specialized ‘penguin-type’ melanosomes (Li *et al.* 2012). Experimental maturation of feathers suggests that melanosomes undergo up to 10–20% taphonomic shrinkage (McNamara *et al.* 2013; Colleary *et al.* 2015). We therefore increased the length and width values of three-dimensionally preserved melanosomes in CUGB P1202 (sample 62; Peteya *et al.* 2016, fig. S2) by 10% and 20% to incorporate this range of potential taphonomic reduction in size.

Raman spectroscopy

We used Raman spectroscopy to assess the presence of melanin in each sample. This is a standard technique for qualitatively assessing the chemical structure of a sample by generating and recording vibrations of molecular bonds on the sample surface (Colthup *et al.* 1990). Raman spectra were generated using a LabRAM High Resolution Raman microscope (HORIBA Scientific; <http://www.horiba.com/uk/scientific/>) with a 50 mW 532 nm excitation laser through a 50 $\times$ objective. We used a wafer of pure silicon to calibrate the instrument, then photobleached each sample using a 10% filter for 10 min prior to data collection. Photobleaching is common technique in which a highly-fluorescent sample is irradiated by a laser with a low wavelength (usually 532 nm) prior to data collection to decrease fluorescent signals (Golcuk *et al.* 2006). Raman spectra were collected at least three times from each sample to account for repeatability of Raman peaks using a grating of 1200 lines per mm, a 400 μ m pinhole, a 100 μ m slit aperture, and a wavenumber range of 300–2500 cm^{-1} . A five second exposure time $\times$ one acquisition was used for each spectrum collection.

These results were compared to spectra of melanin extracted from extant avian feathers: iridescent feathers of a mallard duck (*Anas platyrhynchos*) and a wild turkey (*Meleagris gallopavo*), black feathers of a chicken (*Gallus gallus*) and a red-winged blackbird (*Agelaius phoeniceus*), and brown feathers from a Cooper’s hawk (*Accipiter cooperii*) and a house wren (*Troglodytes aedon*). Melanin was

extracted from the β -keratin matrix using a standard Proteinase-K-based technique (Liu *et al.* 2003).

We collected spectra from an unpigmented remex from the sulfur-crested cockatoo *Cacatua galerita* as a standard for B-keratin and from carbon black as a control for the potential effect of carbon on the surface of the fossil samples. We also removed 2 mm samples from the cuticle preserved on the basal midrib of a *Platanus* leaf (PTRM #20639) from the Late Cretaceous Hell Creek Formation, Mud Buttes (PTRM #P88002), Bowman County, North Dakota, and a carbonized area on a Carboniferous lycopsid root *Stigmara* (CMNH P-21773) from the Joggins Formation, Joggins, Nova Scotia, for comparison to the CUGB P1202 samples as negative controls for melanin. We further collected spectra from a colony of the common feather-degrading (keratinolytic) bacterium *Bacillus licheniformis* to test the potential affinity of preserved granules in CUGB P1202 to microbes (see Moyer *et al.* 2014). The colony of *B. licheniformis* was transferred from tryptic soy agar plates to a sterile unpigmented Mallard nape feather and grown overnight at 37°C. Spectra from extant samples, carbon black, fossil plants and bacteria were generated in a similar manner to the fossil samples, although they were not photobleached to avoid burning the samples. Both fossil and extant samples were examined for burning directly after spectra were collected. If a sample was burned, the resulting spectra were discarded. We used Origin Pro (OriginLab; <http://originlab.com>) to normalize the spectra for comparison and Igor Pro (Wavemetrics; <https://www.wavemetrics.com/>) to conduct peak fitting for each spectrum using a Gaussian distribution.

Institutional abbreviations. BMNHC, Beijing Museum of Natural History, Beijing, China; CMNH, Cleveland Museum of Natural History, Cleveland, Ohio; CUGB, China University of Geosciences, Beijing, China; IVPP, Institute of Vertebrate Paleontology and Paleoanthropology, Beijing, China; PTRM, Pioneer Trails Regional Museum, Bowman, North Dakota.

SYSTEMATIC PALAEONTOLOGY

AVES Linnaeus, 1758

AVIALE Gauthier, 1986 *sensu* Gauthier & de Queiroz, 2001

ORNITHORACES Chiappe, 1995

ENANTIORNITHES Walker, 1981

BOHAIORNITHIDAE Wang *et al.*, 2014a

Gen. et sp. indet

Specimen. CUGB P1202 consists of a nearly complete articulated specimen preserved in part and counterpart with well-preserved melanized feathers (Figs 1, 2). The matrix is dark grey in colour and white material

surrounds the skeleton. The bones and preserved feathers are black in colour. Most of the skeletal elements are preserved in the primary slab (Fig. 1A). With the exception of the skull, which is preserved in oblique lateral view, the skeleton is preserved in ventral view.

Ontogenetic status. The specimen is interpreted as a sub-adult individual. While it has a pair of elongate tail feathers that suggest that the animal had reached sexual maturity at the time of death, some pitting is visible on the periosteal surface of the tibia, suggesting that it had not yet reached skeletal maturity (Chiappe *et al.* 2007). Fusion of the distal carpals to the metacarpals and of the proximal tarsals to the tibia is absent while these states are variably present in other parts of the Bohaiornithidae (fused in the *Zhouornis* holotype and *Bohaiornis* IVPP V 17963; unfused in the *Bohaiornis* holotype, *Shenqiornis*, *Sulcavis*, *Zhouornis* BMNHC Ph756, *Parabohaiornis* and *Longsunguis*) and more basal avialans. CUGB P1202 is approximately half the size of inferred adult bohaiornithid specimens, including the *Zhouornis* holotype and the referred specimen of *Bohaiornis* (Zhang *et al.* 2013; Li *et al.* 2014a) and metacarpal III is only about 61% of the size of *Parabohaiornis* IVPP V 18691, the previous smallest bohaiornithid (45% *Zhouornis*, 49% *Sulcavis*, 51% *Shenqiornis*, 56% *Longsunguis* and *Bohaiornis* IVPP V 17963; Wang *et al.* 2014a). CUGB P1202 is therefore the smallest and potentially the youngest bohaiornithid specimen known to date.

Occurrence. Early Cretaceous Jiufotang Formation, near Lamadong Village, Jianchang County, Liaoning Province, China.

Referral to the Bohaiornithidae. Based on Chiappe (2002) and Chiappe & Walker (2002), synapomorphies of Enantiornithes that are recognizable in CUGB P1202 include: a dorsal margin of the furcula that is narrower than the ventral margin; a well-developed furcular apophysis; a concave central margin of the humeral head; metacarpal III extending further distally than metacarpal II; a deep intercondylar sulcus of the tibiotarsus; and metatarsals II and III wider than metatarsal IV. CUGB P1202 is further referred to Bohaiornithidae based on the following shared synapomorphies (Wang *et al.* 2014a): large caudally-recurved teeth that are robust and sub-conical in shape, the rostral-most of which is smaller than the caudal teeth; a mediolaterally-expanded sternal midline; caudally-expanded lateral trabeculae of the sternum; blunt omally-expanded tips of the furcula; an elongate, robust scapular acromion process; and an elongate pedal digit III ungual that measures >40% (60%) of the length of the tarsometatarsus and is the longest pedal digit.

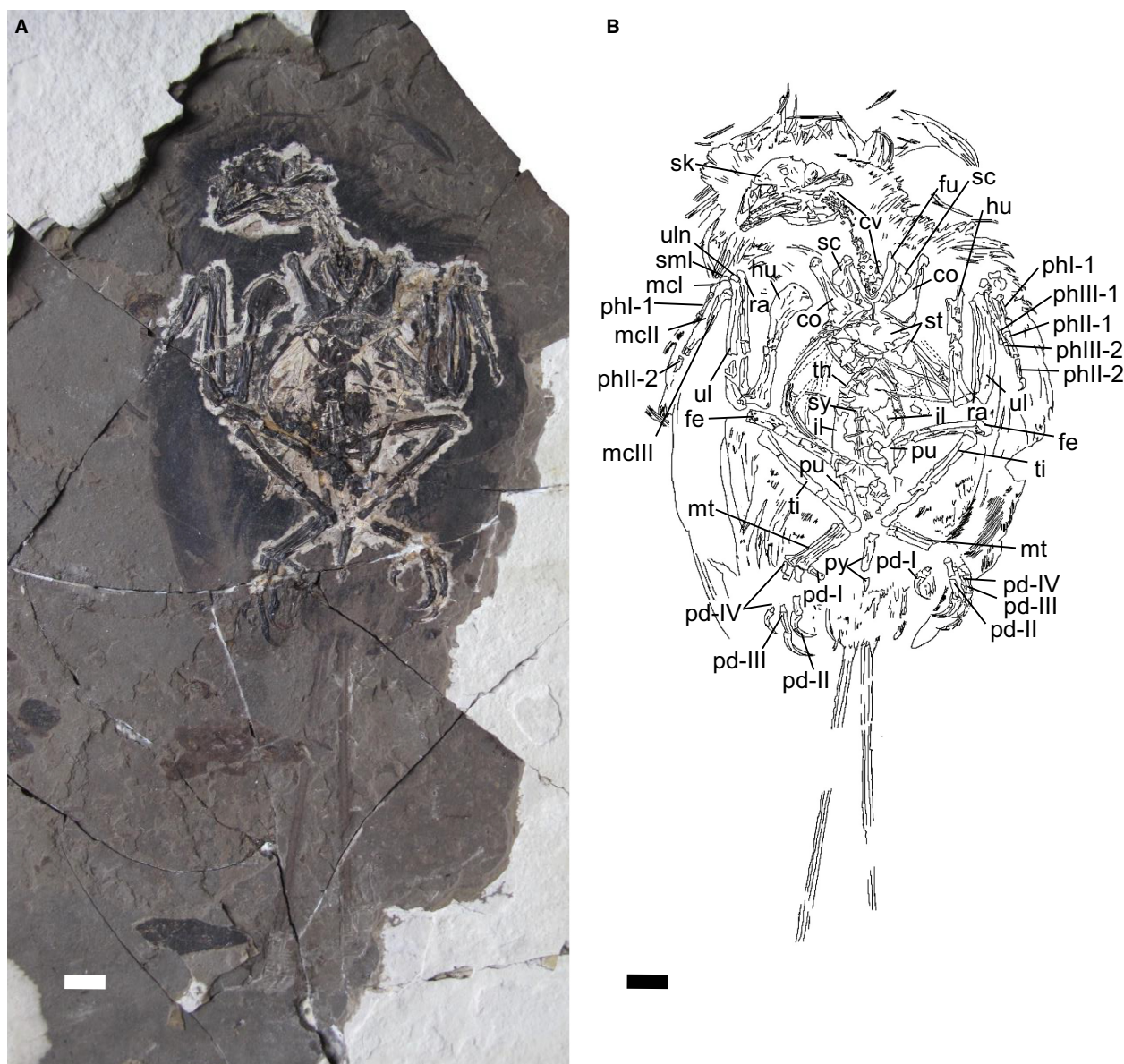


FIG. 1. Ventral view of the new bohaiornithid specimen (CUGB P1202, primary slab). A, photograph of the primary slab. B, interpretive drawing. *Abbreviations:* co, coracoid; cv, cervical vertebrae; fe, femur; fu, furcula; hu, humerus; il, ilium; mcl–III, metacarpals I–III; mt, metatarsals; pd I–IV, pedal digits I–IV; phI–1, first phalanx of digit I; phII/III–1/2, first/second phalanx of digit II/III; pu, pubis; py, pygostyle; ra, radius; sc, scapula; sk, skull; sml, semilunate carpal; st, sternum; sy, synsacrum; th, thoracic vertebrae; ti, tibia; ul, ulna; uln, ulnare. Scale bar represents 1 cm. Colour online.

Preservation. CUGB P1202 consists of a nearly complete skeleton with preserved melanized feathers. A white ‘halo’ surrounds and intermingles with skeletal elements and obscures the proximal ends of feathers. A similar ‘halo’ has previously been described in *Microraptor* sp. and was suggested to be the result of changes in matrix chemistry due to water being trapped between the feathers and the body (Hone *et al.* 2010). In some specimens (e.g. Hone *et al.* 2010) the halo appears to have a gradational contact with the preserved feathers and often occurs in cracks in

the rock itself. In CUGB P1202, this contact is more abrupt and the white material does not occur in cracks in the matrix.

Description

The skull is preserved in left oblique lateral view. The premaxilla contains four robust teeth that begin in the rostral portion of the premaxilla and several neurovascular foramina. Like other

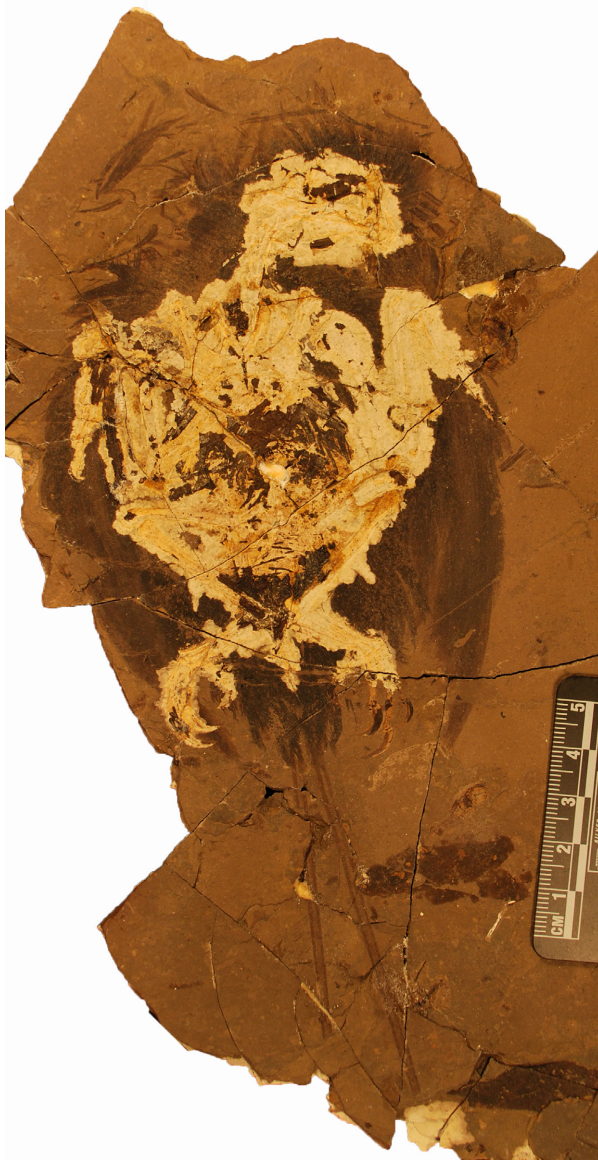


FIG. 2. The new bohaiornithid specimen (CUGB P1202, counter slab). Colour online.

bohaiornithids (Wang *et al.* 2014a), the rostral-most tooth is slightly smaller in size than the other teeth and all of the teeth are unserrated. The nasal process of the premaxilla is elongate and contacts the nasal. The nasal is broad and slightly displaced. The maxilla is damaged caudally. Four robust, conical teeth are preserved in the cranial portion of the maxilla. The maxilla has a relatively long nasal process and has a slender dorsal process that is obscured dorsally by matrix. The frontal has been crushed cranially and obscures the orbit. The jugal is displaced caudally. Like *Bohaiornis*, the jugal is elongate and strap-like, but lacks the slight ventral recurvature that is characteristic of *Bohaiornis* (Li *et al.* 2014a). The jugal bears an elongate caudodorsal process that projects dorsally in a manner similar to *Shenqiornis* and *Bohaiornis* (Wang *et al.* 2010; Li *et al.* 2014a).

The dentaries are straight and unfused rostrally. The rostral portion of the left dentary has been displaced caudodorsally so that it lies directly ventral to the caudal premaxilla. The right dentary shows a visible Meckel's groove, which like *Bohaiornis*, *Longsunguis* and *Parabohaiornis* (Li *et al.* 2014a; Wang *et al.* 2014a) does not extend to its rostral tip. Both dentaries contain at least three robust, slightly recurved, conical teeth in the rostral portion, each of which lack serrations. CUGB P1202 has fewer dentary teeth than any other bohaiornithid. Dental morphology is opposed to *Pengornis*, which has many small teeth that lack recurvature (Zhou *et al.* 2008). Like many other enantiornithines (Chiappe & Walker 2002), the dentary slopes caudoventrally. Caudal elements of the mandible, including the surangular, angular, articular and lateral condyle, are displaced caudally. The surangular has a slightly concave dorsal margin. A process on the dorsal surface of the surangular, interpreted as the lateral mandibular process by Wang *et al.* (2014a), is not preserved on the left surangular, but may be visible on the right surangular when viewed in the x-ray computed tomography image (Fig. 3D). A lateral condyle is preserved on the articular and there may be a short retroarticular process, although the articular is incompletely preserved caudally.

At least eight articulated cervical vertebrae are preserved in ventral view. The cervical vertebrae are approximately as wide as they are long and the cranial cervical vertebrae are weakly heterocoelous. As in other enantiornithines, the centra of the cranial cervical vertebrae bear a small ventral process (Chiappe & Walker 2002). The thoracic series is poorly preserved. Three thoracic vertebrae are preserved on the surface of the part slab and at least two more are revealed by CT scanning. The centra bear deep lateral depressions. Ribs and gastralia are disarticulated and preservation is split between the part slab and the counterpart slab. Most of the exposed ribs are preserved as casts. Six gastralia are preserved on the counterpart slab. Little of the synsacrum is visible on the surface of the part slab. The sacrum is comprised of at least seven fused sacral vertebrae. It bears a thin, shallow central groove similar to that of *Zhouornis* (Zhang *et al.* 2014). At least three disarticulated free caudal vertebrae are preserved. As in other basally divergent avialans, two of the caudal vertebrae retain elongate transverse processes. An elongate rod-like pygostyle comprised of at least four vertebrae is displaced caudally from the synsacrum. The pygostyle is approximately 13.06 mm long and is proximally slightly expanded into a T-shape showing the incorporation of laterally projected transverse processes. Distally, the pygostyle gradually constricts in a manner similar to other bohaiornithids (Wang *et al.* 2014a). The distal end is cracked and is displaced caudally from the rest of the pygostyle.

Much of the sternum is preserved in dorsal view on the counter slab (Fig. 4A, B). Only a small portion of the cranial end of the sternum and the distal left caudolateral process are preserved on the part slab in ventral view (Fig. 4C). The cranial section is poorly preserved and reveals little about the cranial morphology of the sternum. The counter slab preserves two elongate caudolateral processes (lateral trabeculae) and two shorter medial trabeculae. The left lateral trabecula is broken and displaced caudolaterally from the rest of the sternum. The distal ends of the lateral trabeculae are caudally-expanded, as in other

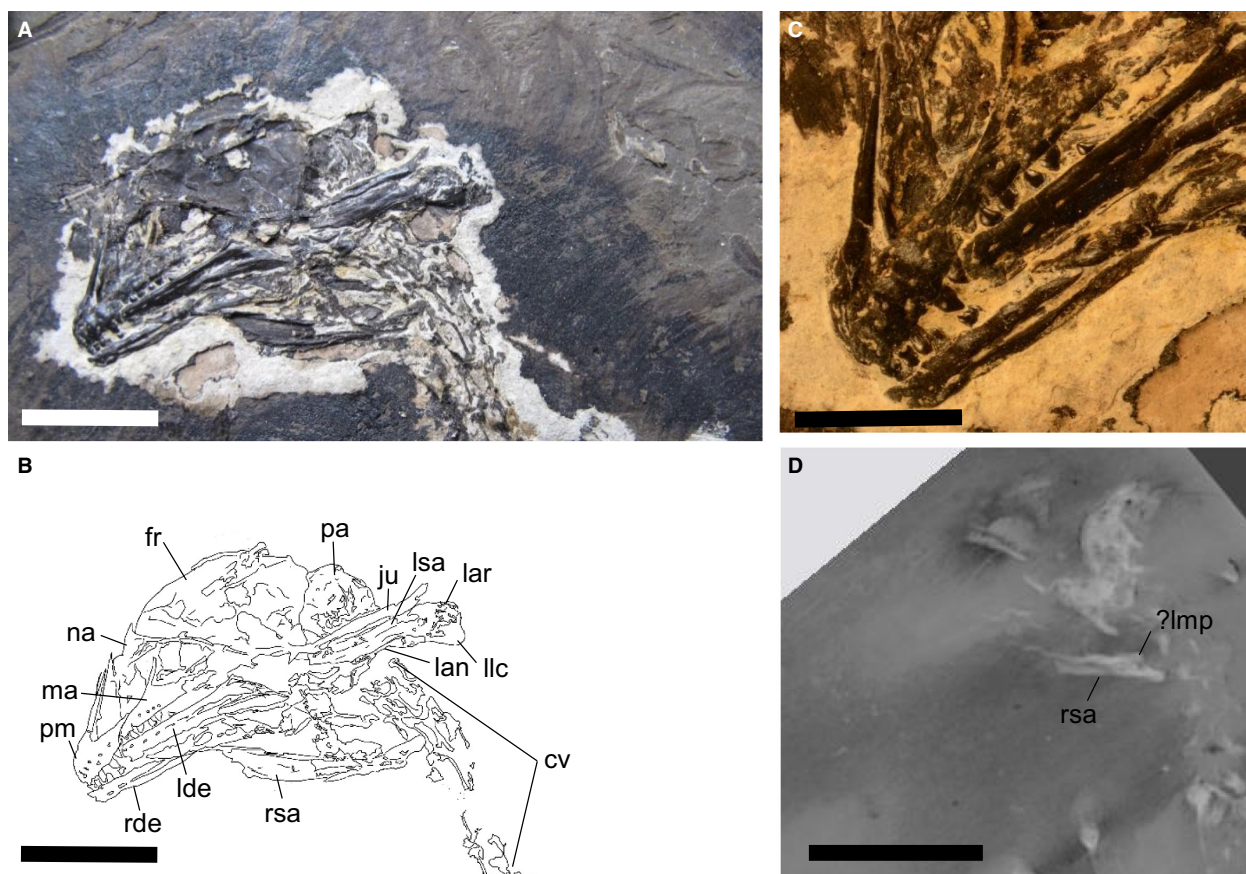


FIG. 3. The skull in CUGB P1202. A, photograph of the skull. B, interpretive drawing. C, details of the dentaries and teeth. D, CT scan of the skull. *Abbreviations:* cv, cervical vertebrae; fr, frontal; ju, jugal; lan, left angular; lar, left articular; lde, left dentary; llc, left lateral condyle; Imp, lateral mandibular process; lsa, left surangular; ma, maxilla; na, nasal; pa, parietal; pm, premaxilla; rde, right dentary; rsa, right surangular. Scale bars represent 1 cm (A, C, D); 0.5 cm (B). Colour online.

bohaiornithids. Similar medial and lateral expansions have been reported for *Zhouornis* (Zhang *et al.* 2013) and *Parabohaiornis* (Wang *et al.* 2014a). The right caudolateral process is displaced from the rest of the sternum and the left caudolateral process is broken distally. The caudal process is subequal in length to the caudolateral processes and the central portion is obscured by a large crack in the rock. A weakly-keeled sternal midline extends to approximately the caudolateral process, which expands caudally into a weak T-shape. A similar morphology has been reported for the referred specimen of *Bohaiornis* (Li *et al.* 2014a).

Both coracoids are preserved in ventral view on the primary slab (Fig. 4). They are also visible in dorsal view the CT images (Fig. 4D). The acrocoracoid process is slightly medially hooked, as in *Bohaiornis* (Li *et al.* 2014a). An enclosed n. supracoracoideus foramen is visible in the CT images (Fig. 4D). The lateral margin of the coracoid is straight to slightly concave. The sternal margin is expanded and slightly concave, as in the referred specimen of *Bohaiornis* (Li *et al.* 2014a). The furcula is robust and generally Y-shaped (Fig. 4C). Its apophysis is long and thin and the furcular rami bear a slight ventral depression as in other enantiornithines. The proximal ends of the scapulae of CUGB P1202 are visible on the surface of the specimen. The

acromion process is greatly extended, robust, and generally rectangular in shape with a slight omal expansion, as in *Bohaiornis*. The scapular blades are strap-like (Fig. 4D).

The forelimb is approximately 80% the total length of the hind limb. The humerus is slightly longer than the ulna and is preserved in cranial view (Fig. 5). Generally, the humerus bears a slight sigmoidal shape. The left humeral head is not preserved, but the right humeral head is visibly strap-like and nearly flat. This morphology is similar to other bohaiornithids, but opposed to the globose humeral head in *Pengornis* and *Xiangornis* (Zhou *et al.* 2008; Hu *et al.* 2012). The deltopectoral crest is broad with a convex dorsal margin and extends along approximately one quarter of the length of the humerus. In *Sulcavis*, the deltopectoral crest extends along more one third of the humerus (O'Connor *et al.* 2013). The deltopectoral crest extends along one third of the humeral length in other bohaiornithids. The distal terminus of the deltopectoral crest is less abrupt than in *Shenqiornis* and *Parabohaiornis* (Wang *et al.* 2010, 2014a).

The right radius and ulna are cracked and damaged. The ulna is significantly shorter than the humerus, which is contrary to the longer ulna preserved in *Sulcavis*, the *Zhouornis* holotype, and *Longusunguis* (O'Connor *et al.* 2013; Zhang *et al.* 2013; Wang *et al.* 2014a). The humerus and ulna are subequal in

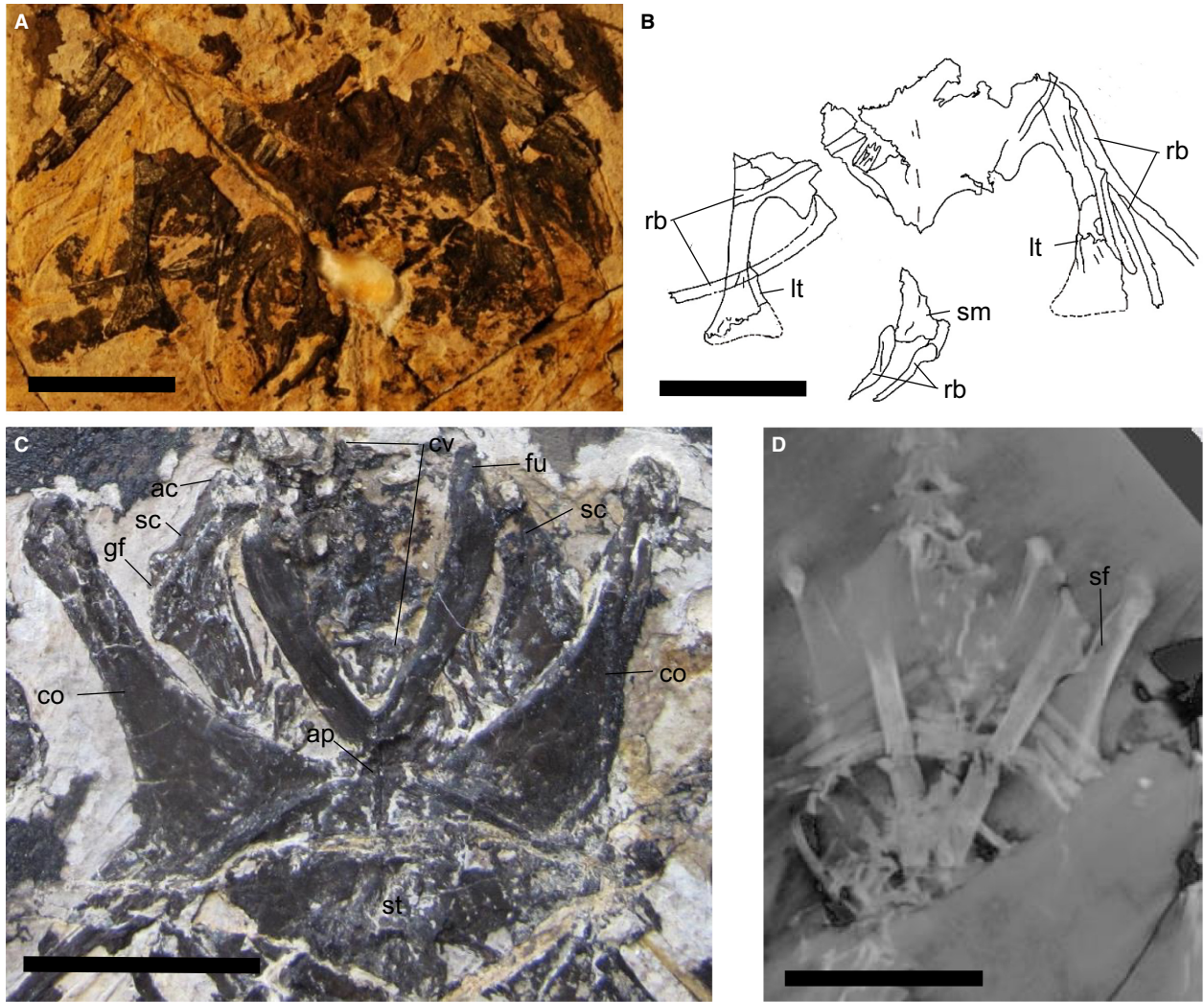


FIG. 4. The pectoral girdle in CUGB P1202. A, photograph of the sternum preserved in dorsal view on the counter slab. B, interpretive drawing of the sternum preserved in dorsal view on the counter slab. C, photograph. D, CT scan of the primary slab, preserved in ventral view. *Abbreviations:* ac, acromion process; ap, furcular apophysis; co, coracoid; cv, cervical vertebrae; fu, furcula; gf, glenoid fossa; lt, lateral trabecula; rb, ribs; sc, scapula; sf, n. supracoracoideus foramen; sm, sternal midline; st, sternum. Scale bars represent 1 cm (A–C); 15 mm (D). Colour online.

length in *Shenqiornis*, *Bohaiornis* and *Parabohaiornis* (Wang *et al.* 2010, 2014a; Li *et al.* 2014a). The left ulna is better preserved and less than twice the width of the radius, unlike *Bohaiornis* (Li *et al.* 2014a). The cranial surface of the radius retains a longitudinal groove that is also present in *Bohaiornis*, *Parabohaiornis*, *Longusunguis*, *Zhouornis* (Wang *et al.* 2014a; Zhang *et al.* 2014) and other Enantiornithes (Chiappe and Walker 2002).

The radiale, ulnare and semilunate carpals are preserved in ventral view on both sides of the specimen and are not fused. The ulnare is heart-shaped, as in other enantiornithines. Metacarpal I is unfused and is short with a broadly rounded cranial margin. Metacarpals II and III appear to be fused proximally but not distally, although damage obscures much of the distal ends of the metacarpals on both sides. Like many other enantiornithines, metacarpal III is longer and extends further distally than metacarpal II. Metacarpal III is slightly curved. Phalanx I-1

is slender and straight. It articulates with a small ungual. Phalanx II-1 is longer and wider than phalanx II-2. Its margins are slightly concave and the general shape is rectangular. Phalanx III-1 is shorter than the other phalanges and does not appear to articulate with an ungual. It is proximally robust and narrows distally. The unguals are small and slightly recurved, as in many other enantiornithines (e.g. other bohaiornithids, *Eoenantiornis* and *Cathayornis*).

The pelvic girdle is incompletely preserved in ventral view on the primary slab and in dorsal view on the counterpart slab, but much of it is visible in CT scans (Fig. 6A–C). Small portions of bone, such as the shaft of the left pubis and partial distal end of the right pubis, and outlines of the pelvic bones are preserved in counterpart. The ilia are preserved on either side of the synsacrum, although the left ilium is better preserved. Like *Parabohaiornis* (Wang *et al.* 2014a), the ilia are not fused to the

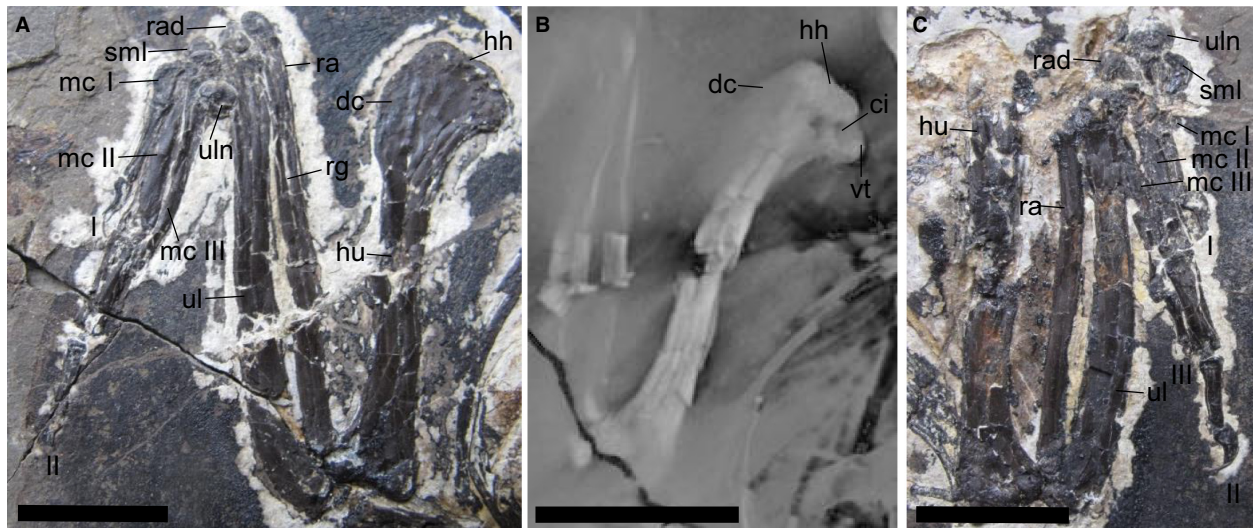


FIG. 5. The forelimbs of CUGB P1202. A, left wing. B, CT scan of the caudal right humerus. C, right wing. *Abbreviations:* ci, capital incisure; dc, deltopectoral crest; hh, humeral head; hu, humerus; I–III, digits I–III; mcI–III, metacarpals I–III; ra, radius; rad, radiale; rg, radial groove; sml, semilunate; ul, ulna; uln, ulnare; vt, ventral tubercle. Scale bars represent 1 cm (A, C); 15 mm (B). Colour online.

sacrum. The proximal ends of the pubes are flared and overlie the proximal femora. The shafts of the pubes are elongate, narrow and bowed slightly. The pubic foot bears a slight distal expansion, as in other bohaiornithids.

Bones of the hind limbs are also generally poorly preserved. The left femur is slightly bowed cranially. At the distal end of the left femur the medial and lateral condyles appear to be separated by a shallow popliteal fossa. Pitting on the periosteal surface of the proximal end of the left tibia suggests that CUGB P1202 is not an adult individual (Fig. 6D). The proximal tarsals are preserved on the right hind limb (Fig. 6E–G). The proximal end of the ascending process of the astragalus is not completely fused to the tibia. In CT images the distal tarsals are visibly not fused to the metatarsals (Fig. 6F).

Breakage obscures the distal ends of the metatarsals on the left, but these are well-preserved on the right side (Fig. 6F, G). Metatarsal II is broader than metatarsal IV, as in other enantiornithines (Chiappe and Walker 2002; Zhou & Zhang 2007) and some basal avialans (e.g. *Asparavis ukhaana*; Clarke & Norell 2002). Metatarsal II and metatarsal IV are subequal in distal extent. The trochlea of metatarsals II and IV are curved slightly plantarly, as in *Bohaiornis*, *Sulcavis*, *Zhouornis* and *Parabohaiornis* (Hu *et al.* 2011; O'Connor *et al.* 2013; Zhang *et al.* 2013; Wang *et al.* 2014a). Metatarsal I is broad with a slight J-shape.

Pedal digits on both sides are cracked and incompletely preserved. Moulds of the original positions of these bones are, however, preserved on the counterpart slab. Based on these moulds, digit II is more robust than the other digits, as in other bohaiornithids (Wang *et al.* 2014a). Digit III is longer than the other digits and ungual III is longer and less highly recurved than all other unguals. This morphology is similarly present in other bohaiornithids (Wang *et al.* 2014a). A keratinous sheath is preserved on all unguals in both part and counterpart. The elongate pedal unguals in CUGB P1202 are also consistent with the

interpretation of bohaiornithids as predominately arboreal birds (Wang *et al.* 2014a).

Plumage

Melanized feathers are preserved throughout the body of CUGB P1202. Feathers are poorly preserved proximally and the articulation between the bones and the feathers is masked by a white matrix ‘halo’ that surrounds the skeleton. Distally, feathers are better preserved and individual barbs can often be distinguished. Preserved feathers include pennaceous body and crown contour feathers, facial feathers, primary and secondary wing feathers, alular feathers on the left side, short rectrices, and two elongate central rectrices. Feathers may occur on the hind limbs, but these are indistinguishable from the covert feathers of the tail. Feathers do not occur on the pedal digits, unlike some maniraptoran dinosaurs (e.g. *Anchiornis huxleyi*; Hu *et al.* 2009).

CUGB P1202 retains long crown feathers near the caudal tip of the rostrum (Fig. 7). Similarly elongate crown feathers occur in *Microraptor*, which have been shown to be the result of diagenetic compression of typical skull feathering rather than a projected crest (Li *et al.* 2012). Tiny, narrow feathers are preserved within the antorbital fenestra on the primary slab (Fig. 7C).

As preserved, the wings are folded in toward the body and are not as darkly-coloured as the crown, body contour feathers, and short rectrices (Fig. 8). An alula is visible on the right side of the body (Fig. 8C, D). The presence of an alula suggests that, like some other

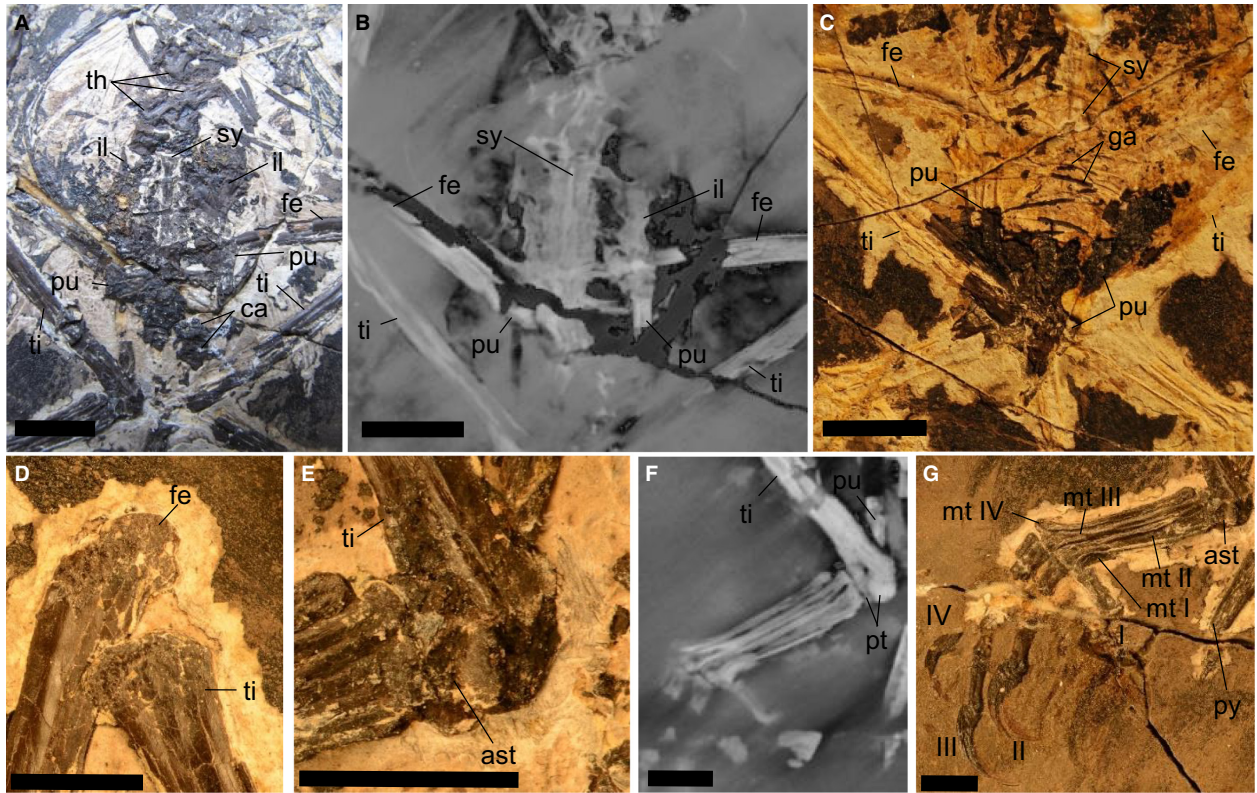


FIG. 6. The pelvic girdle and hind limbs of CUGB P1202. A, photograph of the pelvic girdle preserved on the primary slab. B, CT scan. C, photograph of the pelvic girdle preserved in dorsal view on the counterpart slab. D, periosteal pitting of the left tibia. E, right tarsus. F, CT scan of the right hind limb. G, photograph of the right hind limb. A, B, D–G are in ventral view. *Abbreviations:* ast, astragalus; ca, free caudal vertebrae; fe, femur; ga, gastralia; I–IV, pedal digits I–IV; il, ilium; mt I–IV, metatarsals I–IV; sy, synsacrum; ti, tibia; th, thoracic vertebrae; pt, proximal tarsals; pu, pubis. Scale bars represent 1 cm (A–C); 5 mm (D–G). Colour online.

enantiornithines, CUGB P1202 was capable of manoeuvred flight. This feature is not preserved in any other bohaiornithid, but has been reported for other enantiornithines, including *Eoalulavis*, *Eoenantiornis* and *Grabauornis* (Sanz *et al.* 1996; Zhou *et al.* 2005; Dalsätt *et al.* 2014). Asymmetrically-veined primary feathers are visible and the distal tips of three of these feathers are distinguishable distally on the right side in counterpart (Fig. 8). Four primaries can be distinguished on the left wing. The distal ends of other primary feathers are either not preserved or are obscured by a crack in the matrix. The longest of these feathers is approximately twice the combined length of the humerus and ulna. Two long, veined covert feathers with distinct barbs are preserved projecting from the right side. Individual covert feathers are otherwise not visible on the wings.

Feathers near the tail include at least 12 short (approximately 20 mm), narrow feathers that surround the cranial caudal vertebrae and pygostyle (Fig. 8E). The proximal 86.82 mm of two elongate rectrices are preserved missing their distal tips. These two rectrices appear to be rachis-dominated and lack distinguishable barbs, as in *Confuciusornis*, *Protopteryx* and *Bohaiornis*

(Chiappe *et al.* 1999; Zhang & Zhou 2000; Hu *et al.* 2011).

Elongate granules are preserved both three-dimensionally and as impressions in exceptionally-preserved crown, nape and body contour feathers from CUGB P1202 (Figs 7, 9; samples 62–64). We interpret these granules as melanosomes because their size, shape and chemistry are similar to melanosomes from extant avian feathers. The melanosomes are highly elongate, have high aspect ratios, and many show a degree of organization similar to melanosomes in extant bird feathers (Peteya *et al.* 2016, table S3; Fig. 7E). These melanosomes are only within the size and shape range (Peteya *et al.* 2016, table S3; mean length and aspect ratio, respectively, of sample 62 = 1159.23 nm and 7.13; sample 63 = 1727.36 nm and 5.74; sample 64 = 1628.83 nm and 7.11) of melanosomes from extant iridescent feathers (751.54–1739.82 nm mean length, 3.80–8.57 mean aspect ratio; Li *et al.* 2012). Quadratic canonical discriminant analysis (CDA) suggested that the colour of these preserved feathers was originally iridescent, with greater than 94% probability in each case (sample 62 = 99.86%; sample 63 = 94.17%; sample 64 = 100%; Fig. 10A). These results were not

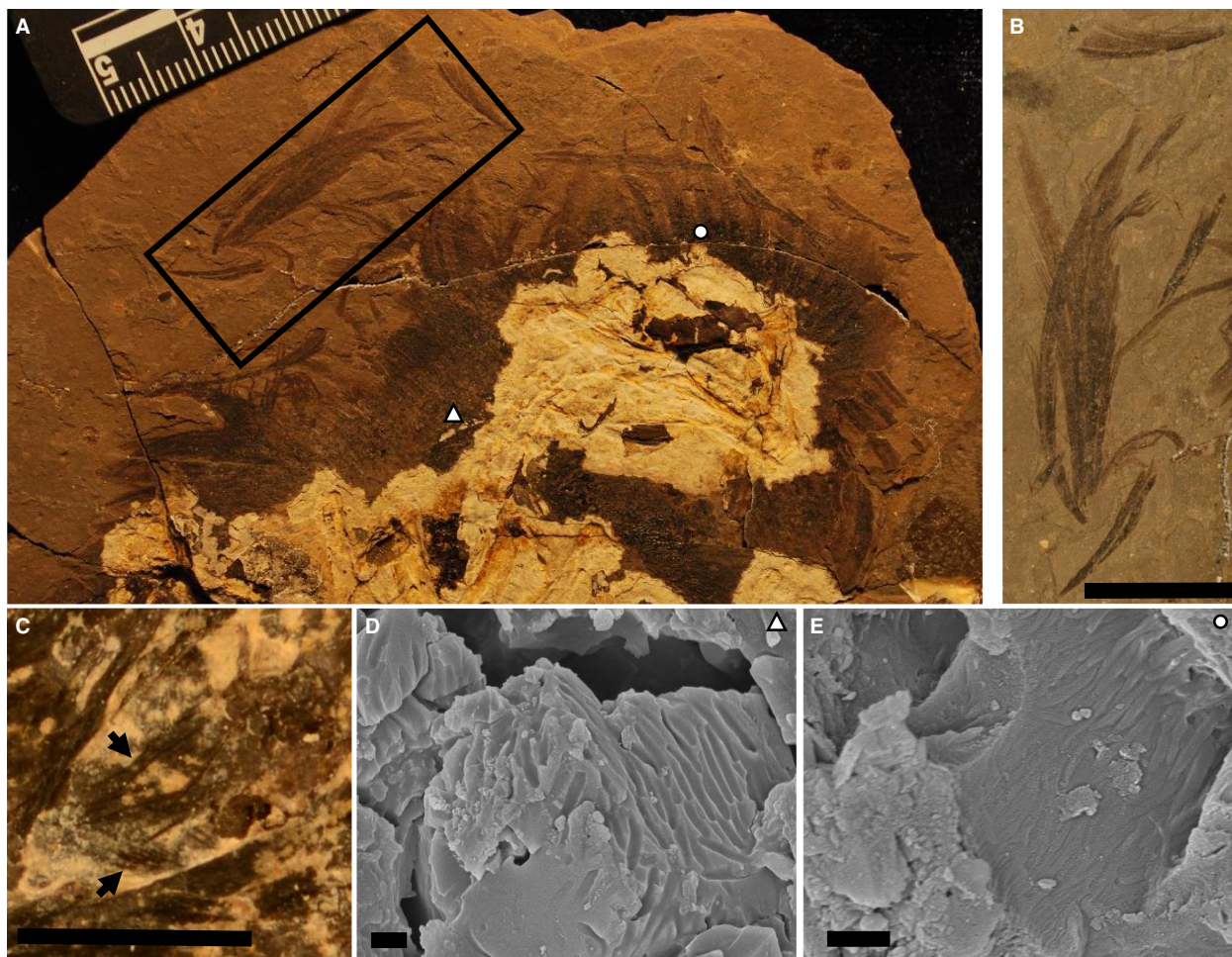


FIG. 7. Crown and neck feathers as well as preserved melanosome morphologies in the new bohaiornithid. A, crown feathers; triangle and circle represent melanosome sampling sites from equivalent points on the primary slab. B, enlargement of area outlined in A; details of loose feathers preserved near the crown. C, hair-like feathers (arrows) within the antorbital fenestra on the primary slab. D, moulds of elongate melanosomes preserved near the base of the neck (triangle on A). E, elongate three-dimensional melanosomes preserved in crown feathers (circle on A). Scale bars represent 1 cm (B); 0.25 cm (C); 1 μm (D, E). Colour online.

significantly affected by an hypothesized 10% to 20% taphonomic shrinkage of three-dimensionally preserved melanosomes, as the probability of iridescent melanosomes in sample 62 remained above 99.82% (Peteya et al. 2016, fig. S2; McNamara *et al.* 2013; Colleary *et al.* 2015). The sample taken from the right wing (sample 65) may preserve degraded melanosomes (Fig. 9C), but these are unmeasurable and therefore excluded from the CDA analysis. Matrix samples do not preserve similar textures or microbodies (Peteya et al. 2016, fig. S3).

Raman spectroscopy suggests that melanin may be preserved in CUGB P1202 feathers, confirming the affinity of the preserved microbodies to melanosomes. Extracted eumelanin from black, brown and iridescent extant bird feathers shows two broad peaks: a smaller Raman peak between 1372 and 1386 cm^{-1} and a taller peak between 1574 and 1585 cm^{-1} . Spectra from all four fossil samples

show two broad peaks: a taller peak between 1566 and 1574 cm^{-1} , and a shorter peak between 1353 and 1364 cm^{-1} (Fig. 10A). These peaks are similar in both location and peak morphology to those generated by extracted eumelanin from extant black, brown and iridescent feathers (Fig. 10A). Eumelanin has previously been reported to have two similar peaks in studies of extracted eumelanin from extant avian feathers and mammalian hair (Galván *et al.* 2013a, b), synthetic eumelanin (Perna *et al.* 2013) and eumelanin from *Sepia officinalis* ink (Huang *et al.* 2004; Centeno & Shamir 2008). Although all melanosomes are likely to contain some ratio of phaeomelanin to eumelanin (McGraw 2006; Liu *et al.* 2014), we could not determine the presence or absence of phaeomelanin within these samples. We also compared the peaks generated by the fossil to carbon black, which produced peaks with a similar wave-number to the fossil spectra, but with a

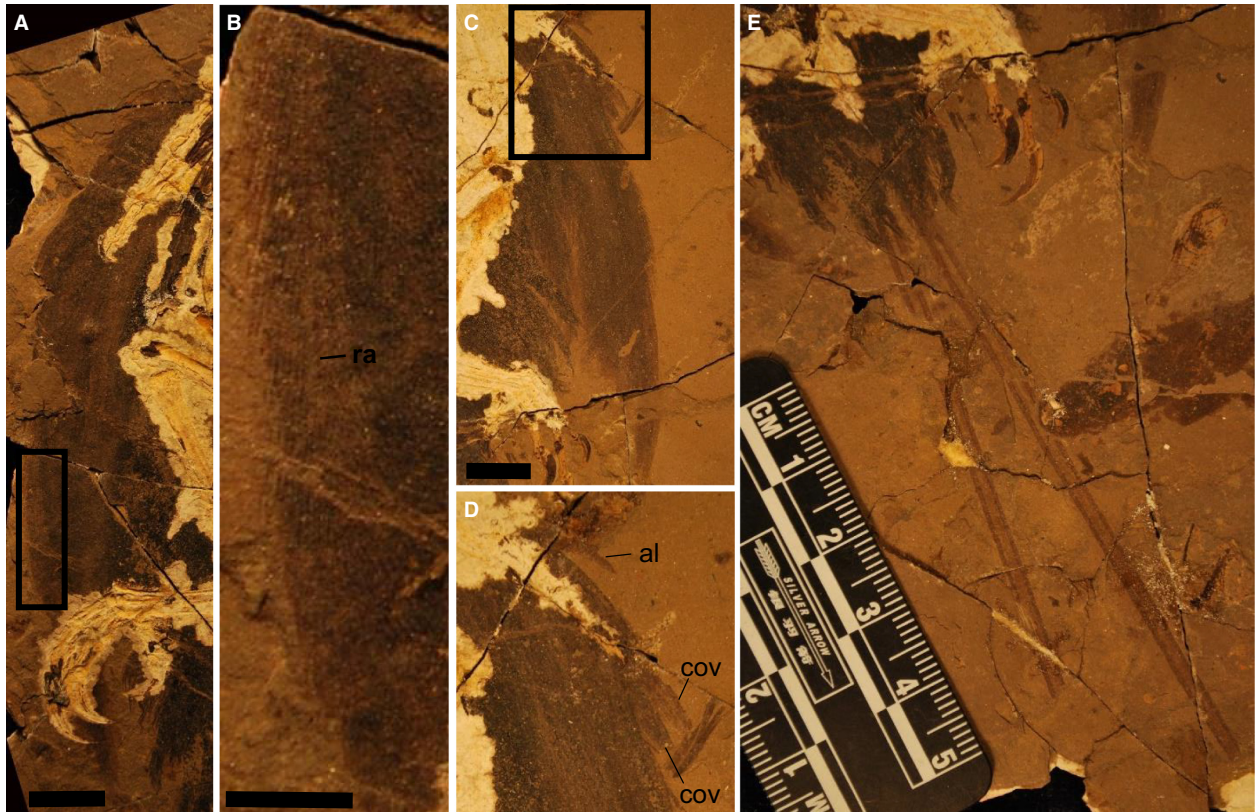


FIG. 8. Details of the remiges and rectrices in the counterpart slab of CUGB P1202. A, the left wing. B, close-up image of the first primary wing feather showing individual barbs and the rachis, outlined in A. C, the right wing. D, details of the alula and coverts preserved on the right wing, outlined in C. E, short plumaceous rectrices and paired elongate rachis-dominated rectrices. Abbreviations: al, alula; cov, covert feathers; ra, rachis. Scale bars represent 1 cm (A, C); 0.25 cm (B). Colour online.

different peak morphology. Carbon peaks are thinner than fossil and extant eumelanin peaks, and the 1360 cm^{-1} peak produced by carbon has a similar intensity to the 1590 cm^{-1} peak (Peteya et al. 2016, fig. S3a). Eumelanin spectra are also not present in *Bacillus licheniformis*, the unpigmented white feather sample, or the matrix samples (Peteya et al. 2016, fig. S3). However, the *Platanus* and *Stigmara* samples presented Raman spectra that are similar to the fossil samples in both peak morphology and location (Peteya et al. 2016, fig. S4). We therefore caution that Raman spectroscopy alone is not a definitive test for the presence of melanin in fossil samples. However, coupled with morphological evidence of preserved melanosomes from scanning electron microscopy and a eumelanin signal that is limited to the preserved plumage, Raman spectroscopy serves as an effective indicator for the preservation of eumelanin in CUGB P1202.

DISCUSSION

CUGB P1202 can be differentiated from other bohaiornithids by a suite of skeletal characteristics, such as a

strap-like jugal bar that lacks ventral recurvature (present in *Bohaiornis* and *Longsunguis*) and has a dorsally-deflected caudal end (dorsal deflection absent in *Zhouornis* and *Parabohaiornis*) that is not forked (it is forked in *Longsunguis* and *Shenqiornis*; *sensu* Wang et al. 2014a); a deltopectoral crest that extends along approximately one-quarter of the length of the humerus (greater than one-third in *Sulcavis*; approximately one-third of the length of the humerus in other bohaiornithids) with a gradual distal terminus (abrupt in *Sulcavis*, *Shenqiornis* and *Parabohaiornis*); a humerus that is longer than the ulna (subequal in length in *Bohaiornis*, *Parabohaiornis* and *Shenqiornis*; humerus shorter in *Sulcavis*, the *Zhouornis* holotype and *Longsunguis*). However, because ontogenetic variation in these features has not been addressed, referral to a specific species in the Bohaiornithidae is not presently supported.

The new specimen presents a combination of juvenile skeletal features and display-linked plumage characteristics that inform the role of sexual selection in Enantiornithes. Currently, there is comparatively little evidence available for ontogenetic shifts in morphology within the Bohaiornithidae and CUGB P1202 is the smallest and

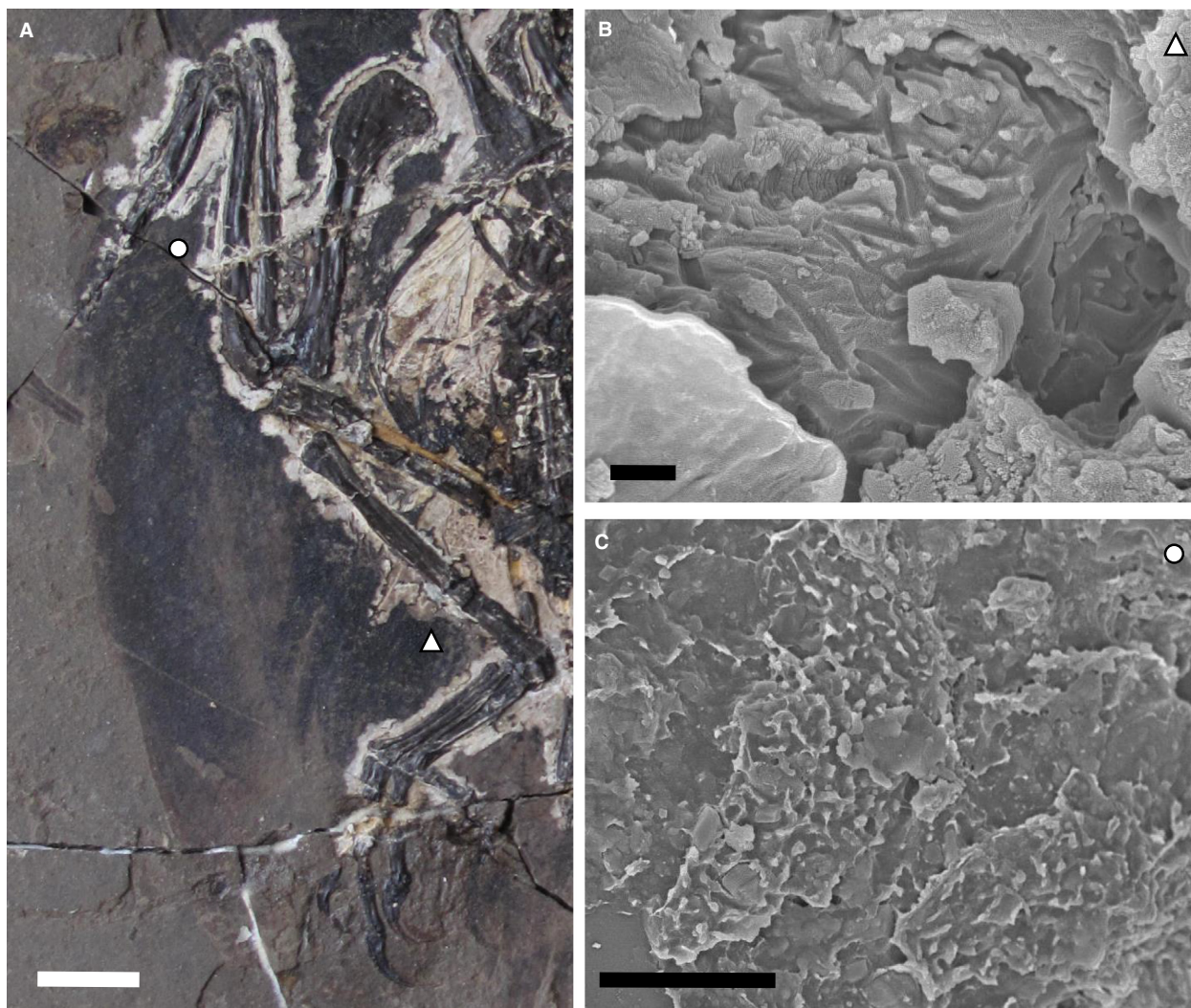


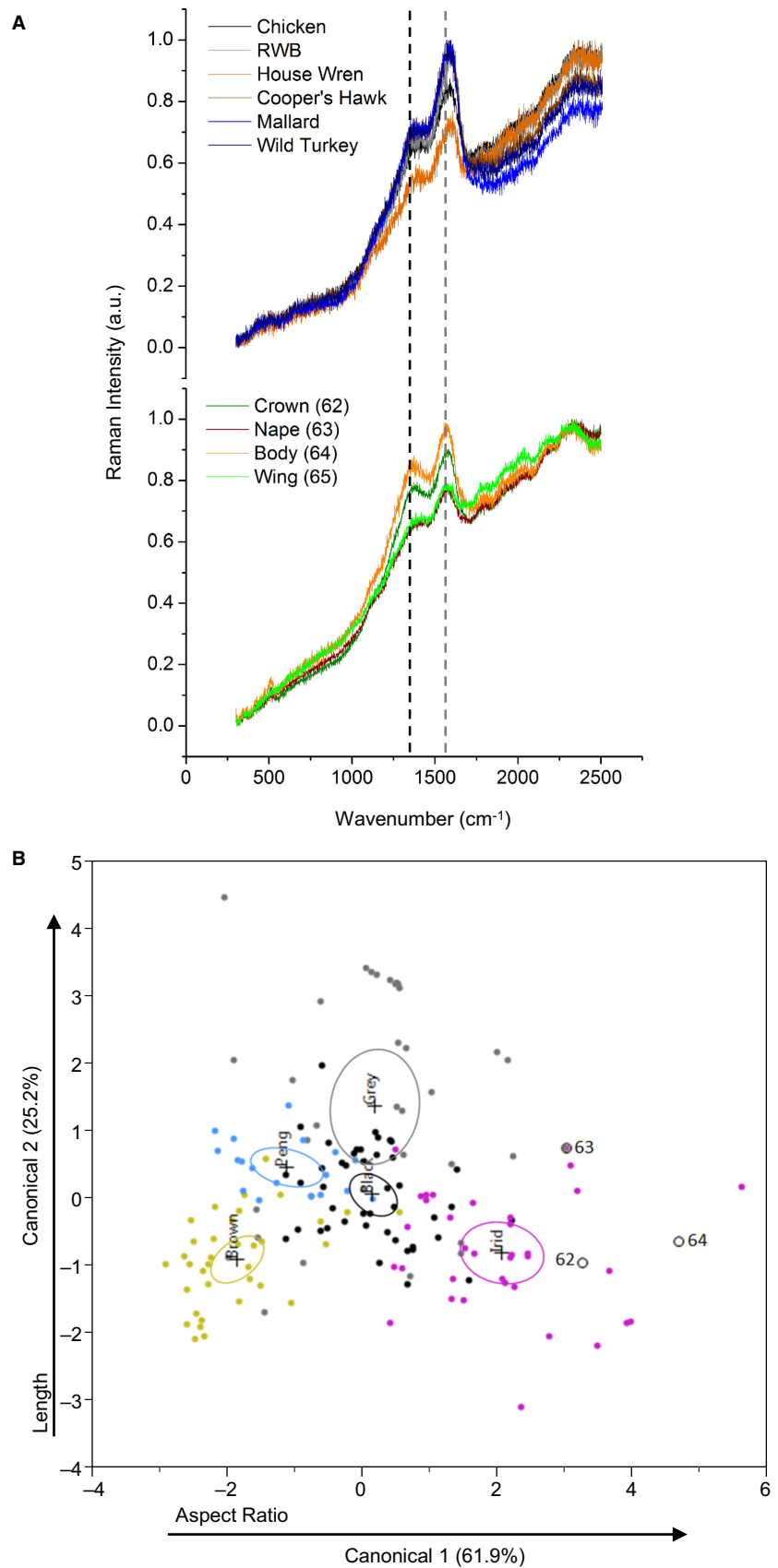
FIG. 9. Right wing and body contour feathers on the primary slab of CUGB P1202. A, photograph of the right wing and long contour feathers next to the tibia; triangle and circle represent melanosome sampling sites. B, moulds of elongate melanosomes preserved near in contour feathers near the tibia (triangle). C, potential degraded melanosomes preserved in the right wing (circle). Scale bars represent 1 cm (A); 1 μ m (B, C). Colour online.

possibly the youngest bohaiornithid yet described. The holotype of *Zhouornis* and the referred specimen of *Bohaiornis* are inferred to be adult specimens based on their large size and complete fusion of compound bones, including the carpometacarpus, tibiotarsus and tarsometatarsus (Li *et al.* 2014a; Wang *et al.* 2014a). CUGB P1202 is approximately half the size of both adult bohaiornithid specimens and only about 75% of the size of the previous smallest specimen (*Parabohaiornis* IVPP V 18691) based on the length of the long bones. Like other subadult bohaiornithids, including *Shenqiornis*, *Sulcavis*, *Longusunguis*, *Parabohaiornis* and *Zhouornis* BMNHC Ph 756 (Wang *et al.* 2010; O'Connor *et al.* 2013; Wang *et al.* 2014a; Zhang *et al.* 2014), CUGB P1202 lacks fusion of compound elements; a state which is considered to be

indicative of subadult ontogenetic stage (e.g. Chiappe *et al.* 2007). Unlike these other subadult bohaiornithids, CUGB P1202 retains some periosteal pitting that would be characteristic of a markedly juvenile animal, although the large, well-developed sternum and plumage indicates that CUGB P1202 was volant and sexually mature (Chiappe *et al.* 2007).

With a mixture of juvenile and subadult skeletal characteristics, CUGB P1202 preserves feather morphologies indicative of sexual maturity. The new specimen possesses well-developed pennaceous remiges, which have been proposed to have developed long before skeletal maturity in juvenile enantiornithines (Chiappe *et al.* 2007; de Souza Carvalho *et al.* 2015). The tail plumage is generally similar to that of other bohaiornithids and *Eoenantiornis* in

FIG. 10. Statistical analysis and Raman spectra for fossil and extant samples. A, Raman spectra for extracted melanin from extant birds (top) and CUGB P1202 samples (bottom). The black dashed line shows the position of a shorter peak at approximately 1370 cm^{-1} . The grey dashed line denotes the position of a taller peak at approximately 1570 cm^{-1} . B, quadratic canonical discriminant analysis of melanosome morphology. Open circles and numbers refer to CUGB P1202 samples. Coloured points refer to average feather melanosome morphologies from extant avian species (Li *et al.* 2012, database S1). Black points are black, grey points are grey, brown points are red-brown, purple points are iridescent, and blue points are penguin-type melanosome morphologies. Ellipsoids represent the 95% confidence interval. *Abbreviations:* Irid, iridescent; Peng, penguin-type; RWB, red-winged blackbird. Colour online.



that short rectricial feathers appear dominantly plumaceous rather than pennaceous. However, CUGB P1202 and the *Bohaiornis* holotype (Hu *et al.* 2011) are the only known bohaiornithids in which paired and elongate, rachis-dominated rectrices are preserved, although neither preserves the distal tips of these feathers. In other basal birds, this condition is variable. *Confuciusornis* and the primitive enantiornithine *Protopteryx* similarly preserve paired rectrices with thick rachises (Chiappe *et al.* 1999; Zhang & Zhou 2000, Clarke *et al.* 2006) while some pengornithids preserve elongate, fully pennaceous rectrices (Hu *et al.* 2014, 2015; Wang *et al.* 2014b) or are proposed to exhibit fan-shaped morphologies (O'Connor *et al.* 2015). The rachis-dominated morphology has been interpreted as 'ribbon-like' or 'streamer-like' by some authors (e.g. Chiappe *et al.* 1999; Li *et al.* 2012) while others suggest that since these feathers seem to undergo little post-burial deformation, they were more likely to have been stiff structures (Wang *et al.* 2014b). The elongate tail feathers preserved in nearly three-dimensions in *Cratoavis* support a rigid morphology (de Souza Carvalho *et al.* 2015). In extant birds, elongate tail feathers are often used as sexual ornaments. For example, elongate tail feathers in male widowbirds and scarlet-tufted malachite sunbirds have been shown to be more favourable to females than shorter rectrices (Andersson 1982, 1992; Evans & Hatchwell 1992). The highly-modified rectrices preserved in the new specimen and in other extinct maniraptoran dinosaurs may have similarly functioned in intraspecific communications, specifically ornamentation (Clarke *et al.* 2006; Li *et al.* 2012; Wang *et al.* 2014b; de Souza Carvalho *et al.* 2015). The presence of elongate rectrices has been proposed to be potentially limited to males in some basal birds. Sexual dimorphism has been hypothesized for *Confuciusornis* due to the presence of paired rectrices on some specimens and lack of these rectrices on others (Hou *et al.* 1999; Chiappe *et al.* 1999, 2008) and by the proposed presence of medullary bone in a female *Confuciusornis* that lacks elongate rectrices and its absence in purported males with these feathers (Chinsamy *et al.* 2013). Additional sampling is necessary, but it is possible that the presence of similar paired streamers in CUGB P1202 and *Bohaiornis* is associated with sexual dimorphism in bohaiornithids.

The morphology of melanosomes preserved in the feathers of CUGB P1202 is consistent with a role for colour in sexual communication in enantiornithines. The melanosomes are highly elongate and there is a slight degree of parallel organization of the melanosomes similar to the weakly-iridescent melanosomes of some extant bird feathers (Fig. 6; Li *et al.* 2012). In extant feathers, iridescence is produced by the coherent scattering of light through nanostructures of differing refractive indices, specifically layers of β -keratin, melanin and air. A layer of

elongate melanosomes below these nanostructural layers serves to absorb incident light and saturate the colour that is produced (Prum 2006). Highly elongate melanosomes associated with iridescence have previously been reported for fossil maniraptoran dinosaur feathers (Vinther *et al.* 2010; Li *et al.* 2012), although CUGB P1202 is the first enantiornithine reported to preserve iridescent melanosome morphologies.

As for previously reported fossil iridescent feathers (Vinther *et al.* 2010; Li *et al.* 2012), the specific hue of the iridescent feathers preserved in the new specimen cannot be determined due to the lack of evidence of preserved keratin. In extant birds, iridescent plumage serves a primary function in intraspecific communications, namely sexual selection and display (Doucet & Meadows 2009), and may serve as an honest signal for the condition of a potential mate (McGraw *et al.* 2002). There are, however, other potential functions of iridescence that have yet to be thoroughly characterized, such as startling potential prey or predator animals, other interspecific communications, crypsis or mechanical strengthening of the feather structure (Doucet & Meadows 2009; Maia *et al.* 2012). These other functions therefore cannot be ruled out for the plumage of CUGB P1202, although coupled with ornamental plumage a role for sexual signaling seems likely.

The combination of immature skeletal characteristics and ornaments in CUGB P1202 is indicative of the importance of sexual selection in basal enantiornithines. In extant avian taxa, skeletal maturity typically occurs long before sexual maturity and males sometimes delay the production of sexual ornaments (e.g. long-tailed manakin *Chiroxiphia linearis*; Arévalo & Heeb 2005). However, bone histology of non-avian maniraptoran dinosaurs and the presence of ovarian tissues in skeletally-immature fossil birds, including some enantiornithines and basal birds such as *Jeholornis* and *Confuciusornis*, suggest that like extant crocodiles these animals reached sexual maturity long before skeletal maturity (Erickson *et al.* 2007; Chinsamy *et al.* 2013; Zheng *et al.* 2013; O'Connor *et al.* 2014). CUGB P1202 retains immature skeletal characteristics, including periosteal pitting of the long bones and lack of fusion of compound bones, while also preserving likely sexual ornaments, such as the elongate paired rectrices and iridescent plumage. CUGB P1202 had therefore reached sexual maturity before full developmental maturity.

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DATA ARCHIVING STATEMENT

Data for this study are available in the Dryad Digital Repository: <http://dx.doi.org/10.5061/dryad.cv03k>

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